ANURAG UNIVERSITY | SCHOOL OF ENGINEERING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

R PROGRAMMING
PBL PRESENTATION
ON SAMPLE SUPERSTORE SALES ANALYSIS

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DATASET DESCRIPTION

About the Dataset

It contains transactional sales records from a fictional US superstore, covering orders across different regions, cities, product categories, and customer segments.

The dataset is ideal for exploring sales trends, profitability, geographic distribution of orders, and product-level performance.

Key Facts

No. of Rows: 9,994 observations

No. of Columns: 13 variables

Region: Central, East, South, West

Categories: Furniture, Office Supplies, Technology

Ship Modes: Second Class, Standard Class, etc.

Key Fields: Sales, Quantity, Discount, Profit

Tool Used: R (ggplot2, plotly, treemap, dplyr)

Source: Kaggle – Sample Superstore Dataset

IMPORTING THE DATASET

Loading the CSV file and performing initial exploration using head(), colnames() and str().

```
data <- read.csv("C:\\Users\\Adithi\\Downloads\\archive (2)\\SampleSuperstore.csv")

head(data)
colnames(data)
str(data)
```

```
> data <- read.csv("C:\\Users\\Adithi\\Downloads\\archive (2)\\SampleSuperstore.csv")
>
> head(data)
       Ship.Mode   Segment       Country            City      State Postal.Code Region
1   Second Class  Consumer United States       Henderson   Kentucky       42420  South
2   Second Class  Consumer United States       Henderson   Kentucky       42420  South
3   Second Class Corporate United States     Los Angeles California       90036   West
4 Standard Class  Consumer United States Fort Lauderdale    Florida       33311  South
5 Standard Class  Consumer United States Fort Lauderdale    Florida       33311  South
6 Standard Class  Consumer United States     Los Angeles California       90032   West
         Category Sub.Category    Sales Quantity Discount    Profit
1       Furniture    Bookcases 261.9600        2     0.00   41.9136
2       Furniture       Chairs 731.9400        3     0.00  219.5820
3 Office Supplies       Labels  14.6200        2     0.00    6.8714
4       Furniture       Tables 957.5775        5     0.45 -383.0310
5 Office Supplies      Storage  22.3680        2     0.20    2.5164
6       Furniture  Furnishings  48.8600        7     0.00   14.1694
```

Figure: Dataset structure loaded from .csv with read.csv()

DATASET STRUCTURE (colnames & str)

Using colnames() and str() to explore column names, data types, and sample values.

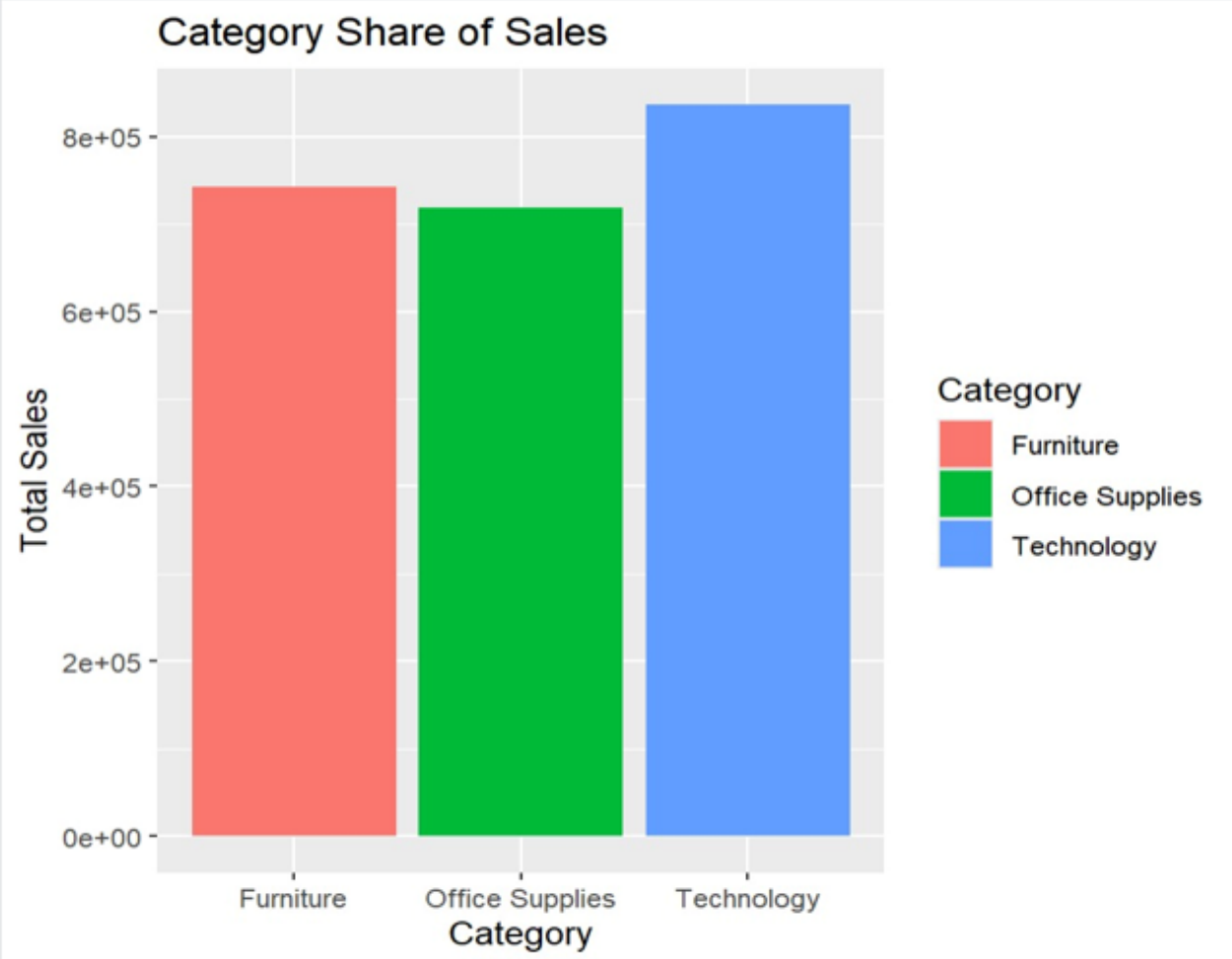
```
> colnames(data)
 [1] "Ship.Mode"    "Segment"      "Country"      "City"         "State"        "Postal.Code"
 [7] "Region"       "Category"     "Sub.Category" "Sales"        "Quantity"     "Discount"
[13] "Profit"
> str(data)
'data.frame':   9994 obs. of  13 variables:
 $ Ship.Mode   : chr  "Second Class" "Second Class" "Second Class" "Standard Class" ...
 $ Segment     : chr  "Consumer" "Consumer" "Corporate" "Consumer" ...
 $ Country     : chr  "United States" "United States" "United States" "United States" ...
 $ City        : chr  "Henderson" "Henderson" "Los Angeles" "Fort Lauderdale" ...
 $ State       : chr  "Kentucky" "Kentucky" "California" "Florida" ...
 $ Postal.Code : int  42420 42420 90036 33311 33311 90032 90032 90032 90032 90032 ...
 $ Region      : chr  "South" "South" "West" "South" ...
 $ Category    : chr  "Furniture" "Furniture" "Office Supplies" "Furniture" ...
 $ Sub.Category: chr  "Bookcases" "Chairs" "Labels" "Tables" ...
 $ Sales       : num  262 731.9 14.6 957.6 22.4 ...
 $ Quantity    : int  2 3 2 5 2 7 4 6 3 5 ...
 $ Discount    : num  0 0 0 0.45 0.2 0 0 0.2 0.2 0 ...
 $ Profit      : num  41.91 219.58 6.87 -383.03 2.52 ...
```

Figure: str(data) output showing 9994 observations across 13 variables

CATEGORY SHARE OF SALES

Bar chart comparing total sales across Furniture, Office Supplies, and Technology categories using ggplot2.

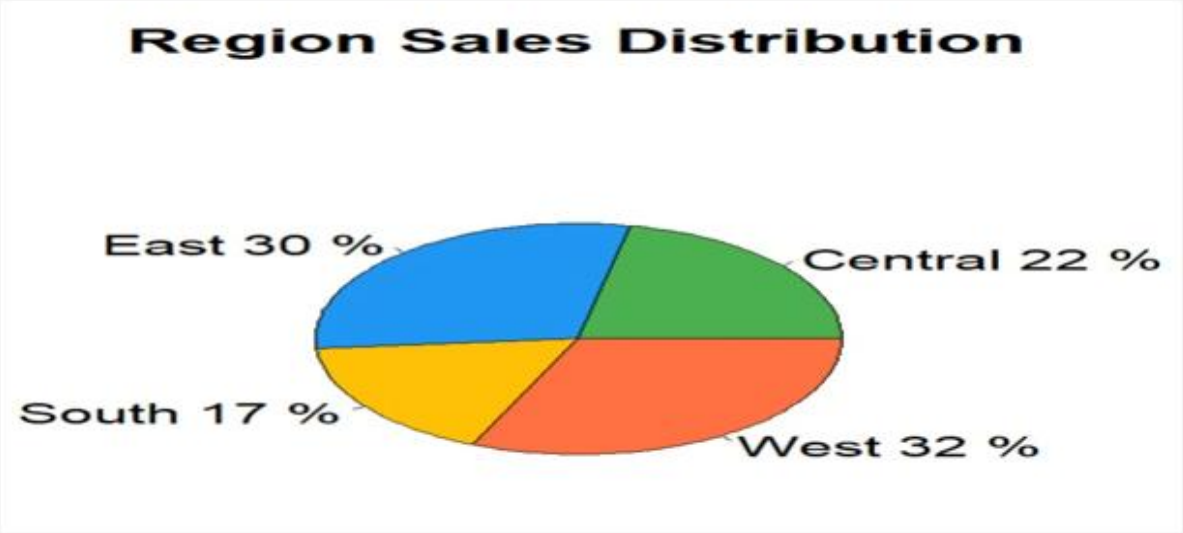
```r
#Category share
category_sales <- data %>%
  group_by(Category) %>%
  summarise(Total_Sales = sum(Sales))


ggplot(category_sales,
       aes(x = Category, y = Total_Sales, fill = Category)) +
  geom_bar(stat = "identity") +
  labs(title = "Category Share of Sales",
       x = "Category",
       y = "Total Sales")
```



Technology leads total sales, followed closely by Furniture and Office Supplies.

REGION SALES DISTRIBUTION (PIE CHART)

Pie chart showing percentage share of total sales by region. West leads with 32%, followed by East at 30%.



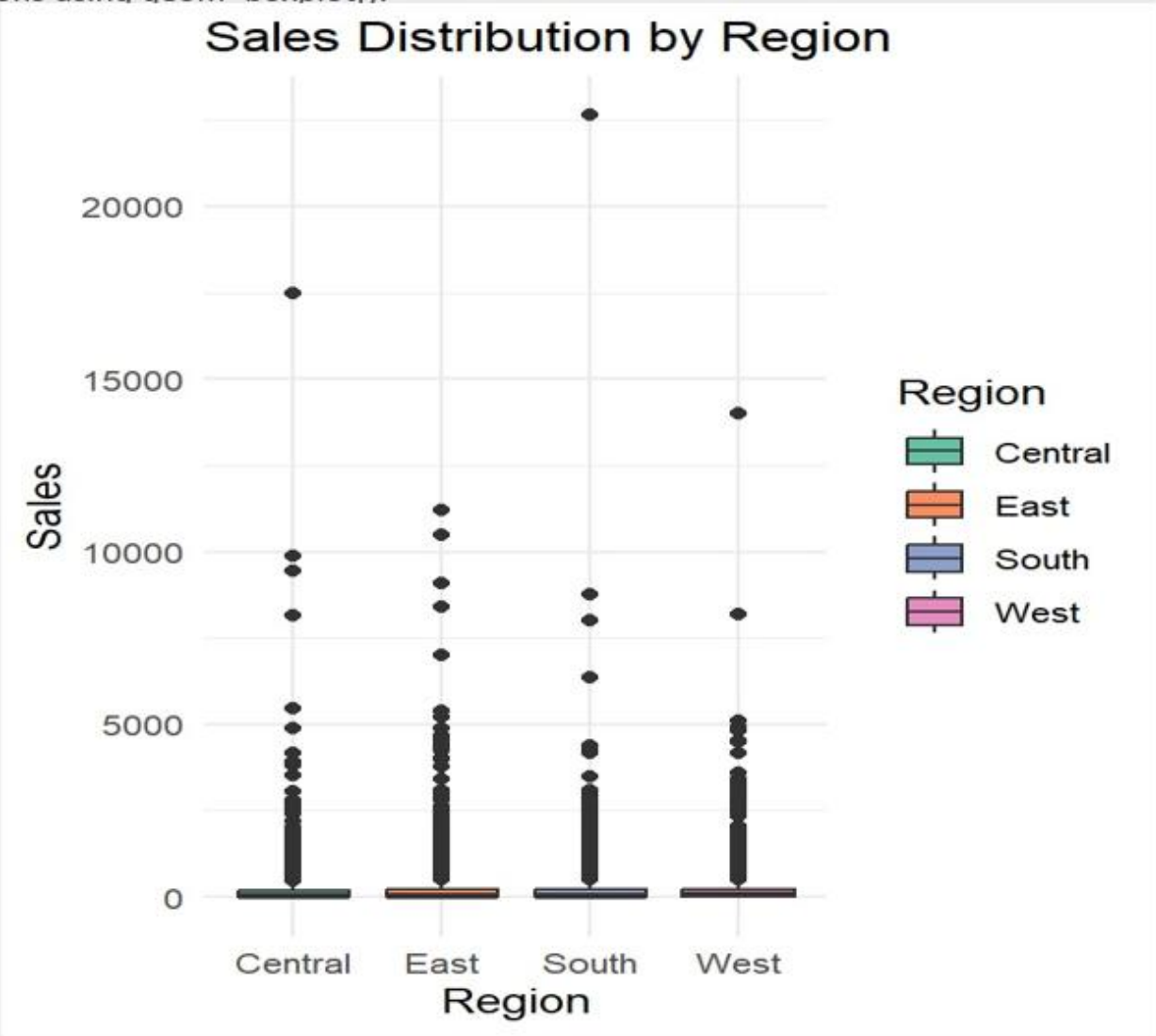
- West: 32%
- East: 30%
- Central: 22%
- South: 17%

```r
# Region Share
region_sales <- data %>%
  group_by(Region) %>%
  summarise(Total_Sales = sum(Sales))
# Calculate percentages
percent <- round(region_sales$Total_Sales / sum(region_sales$Total_Sales) * 100)
# Labels with region + %
labels <- paste(region_sales$Region, percent, "%")

pie(region_sales$Total_Sales,
    labels = labels,
    col = c("#4CAF50", "#2196F3", "#FFC107", "#FF7043"),
    main = "Region Sales Distribution")
```

SALES DISTRIBUTION BY REGION (BOX PLOT)

Box plot showing spread and outliers of Sales values across all four regions using geom_boxplot().

```
#Sales Share
ggplot(data, aes(x = Region, y = Sales, fill = Region)) +
  geom_boxplot() +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Sales Distribution by Region",
       x = "Region",
       y = "Sales") +
  theme_minimal()
```

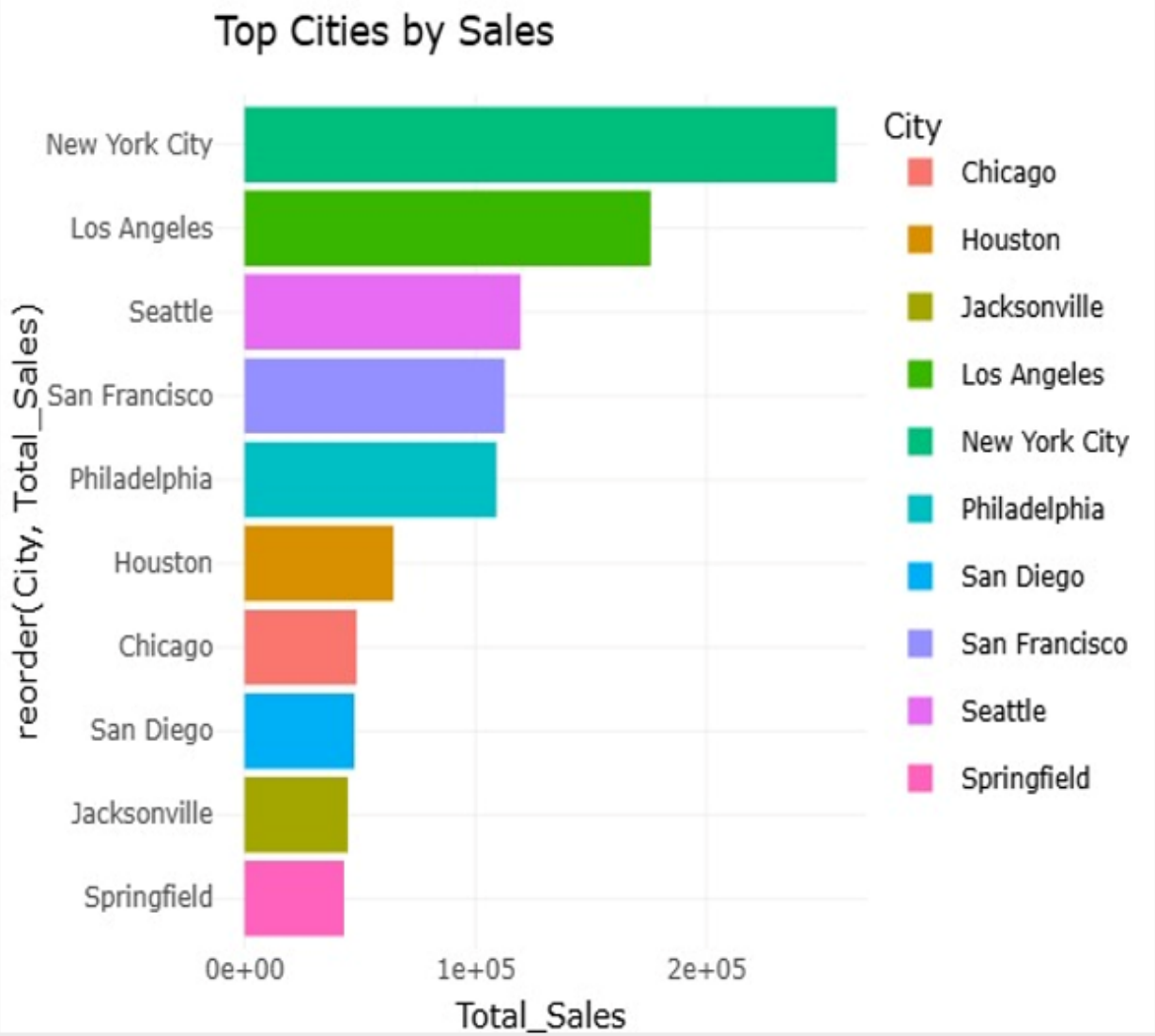


Most sales are concentrated near zero with significant high-value outliers visible across all regions. South has the highest outlier (~22,000).

TOP CITIES BY SALES

Horizontal bar chart using ggplotly() showing the top 10 cities by total sales, arranged in descending order.

```
city_sales <- data %>%
  group_by(City) %>%
  summarise(Total_Sales = sum(Sales)) %>%
  arrange(desc(Total_Sales)) %>%
  head(10)

p <- ggplot(city_sales,
            aes(x = reorder(City, Total_Sales),
                y = Total_Sales,
                fill = City)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  theme_minimal() +
  labs(title = "Top Cities by Sales")

ggplotly(p)
```

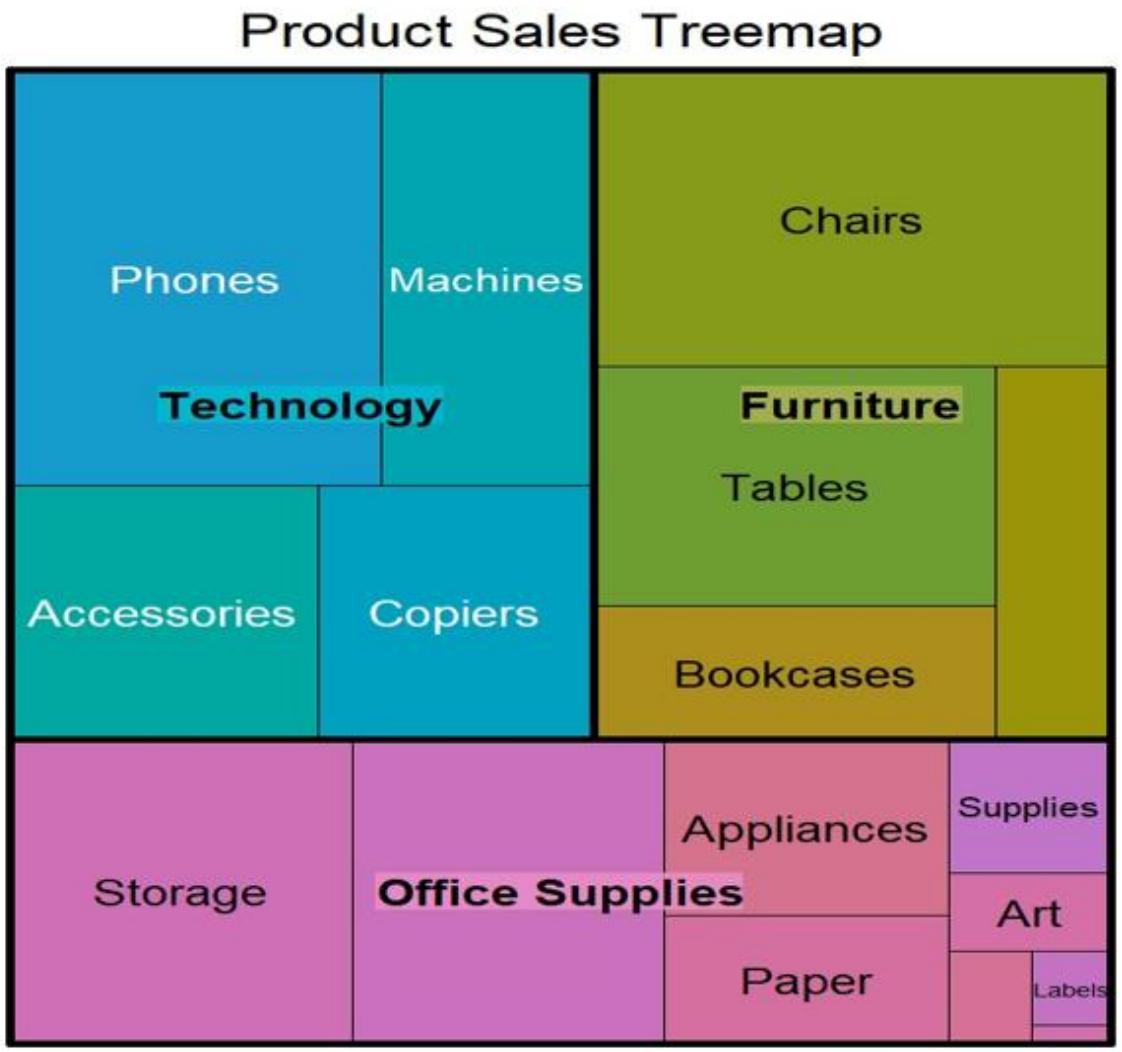


New York City leads with the highest total sales, followed by Los Angeles. Both cities significantly outperform the remaining top 10.

PRODUCT SALES TREEMAP

Hierarchical treemap using the treemap package showing sales by Category and Sub-Category. Tile size ∝ sales volume.

```r
install.packages("plotly")
install.packages("treemap")
library(plotly)
library(treemap)

treemap(data,
        index = c("Category", "Sub.Category"),
        vSize = "Sales",
        title = "Product Sales Treemap")
```

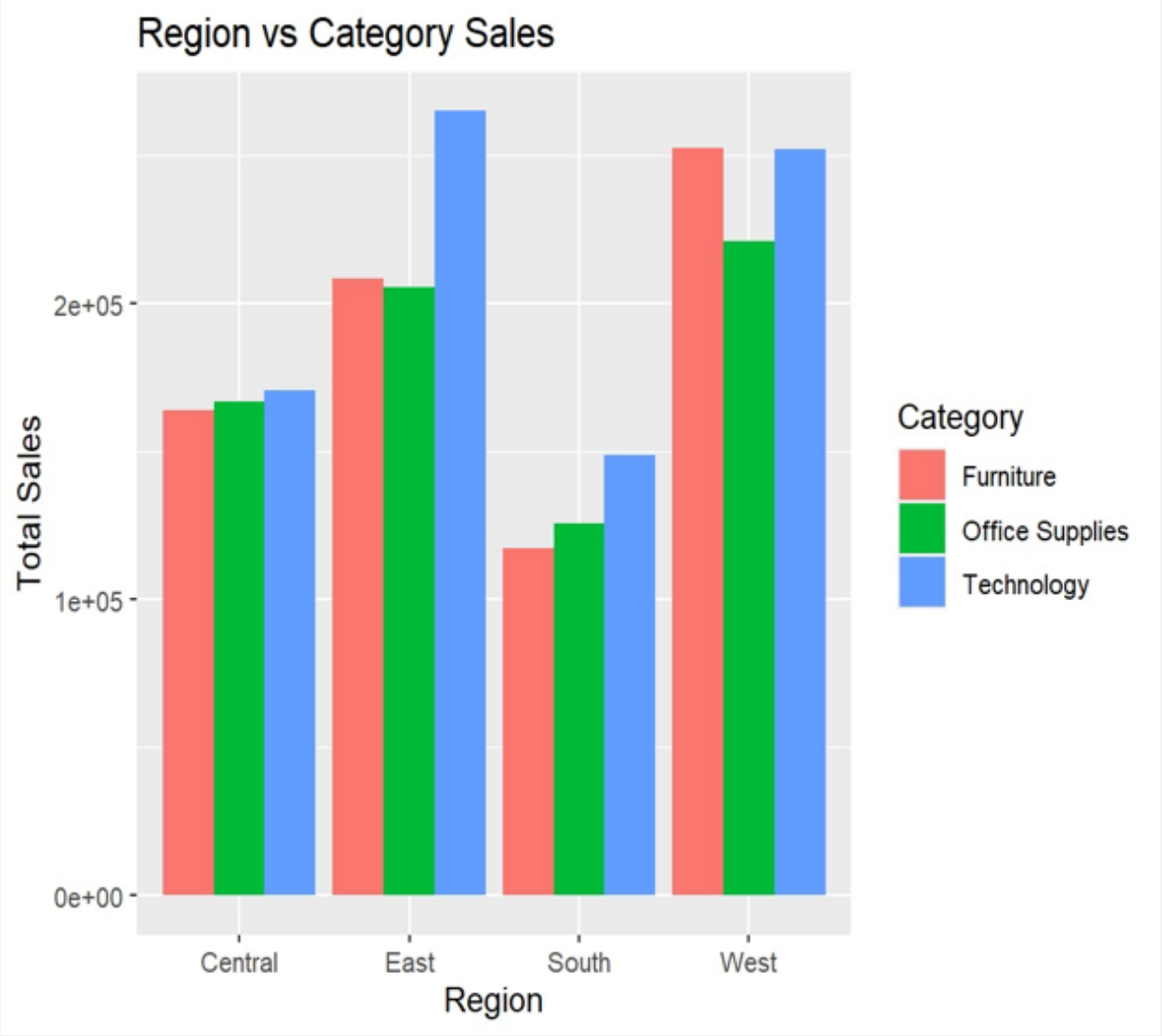


Phones and Chairs are the top sub-categories by sales. Technology dominates the upper-left portion of the treemap.

REGION vs CATEGORY SALES (GROUPED BAR)

Grouped bar chart comparing total sales of each product category across all four regions using geom_bar with dodge.

```r
#Region vs Category Comparison
region_category <- data %>%
  group_by(Region, Category) %>%
  summarise(Total_Sales = sum(Sales))

ggplot(region_category,
       aes(x = Region, y = Total_Sales, fill = Category)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Region vs Category Sales",
       x = "Region",
       y = "Total Sales")
```

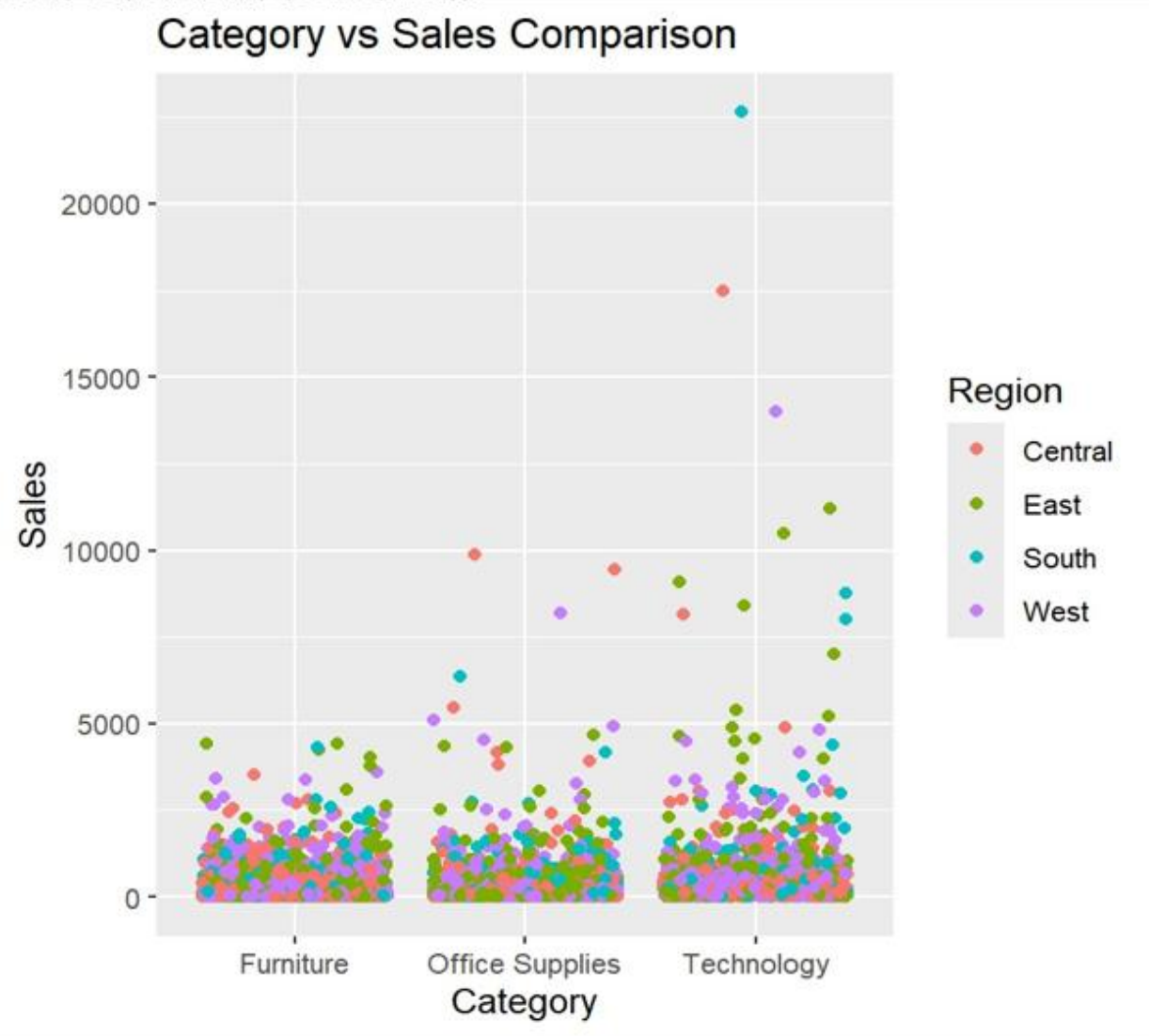


East and West regions lead across all categories. Technology consistently generates the highest sales per region.

CATEGORY vs SALES COMPARISON (SCATTER PLOT)

Jitter scatter plot showing individual sales values by category, color-coded by region using geom_jitter().

```
#Category vs Sales Comparison (Scatter Plot)
ggplot(data,
       aes(x = Category, y = Sales, color = Region)) +
  geom_jitter() +
  labs(title = "Category vs Sales Comparison",
       x = "Category",
       y = "Sales")
```



Technology has the widest sales range with extreme high-value outliers. South region has the highest single sale (~22,638). Most transactions cluster below 2,500.

THANK YOU